41053 Materials and Manufacturing Engineering A Mini Project Group 30

Design, Fabrication and Testing of Vertical Axis Wind Turbine

Members:

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Group Declaration

Revision History Table

Member Name	Contribution	Date	Signature
Nhan Dang	Modelled the components for the wind turbine including the main shaft, half-round PVC blade 20 mm diameter, aluminium tube 60 mm length with three holes to be connected to the PVC blade, Wooden block to hold the short pipe/blades over the shaft, Bearing housing upper half, Bearing housing base through CAD on Solidworks	02/10/2024	N.D
Nhan Dang	Assembled and mated the components together in an Assembly file on Solidworks for the wind turbine	02/10/2024	N.D
Jarrel Redoblado	Mated the components together in the assembly file on Solidworks for the wind turbine	02/10/2024	J.R
Nhan Dang	Participated in the fabrication of the Wind Turbine project	Throughout the fabrication period	N.D

Jarrel Redoblado	Participated in the fabrication of the Wind Turbine project	Throughout the fabrication period	J.R
Hong Nam Hoang	Participated in the fabrication of the Wind Turbine project Record the video of project working	Throughout the fabrication period	H.H
Nathan Matthew Wijaya	Participated in the fabrication of the Wind Turbine project	Throughout the fabrication period	N.W
Nhan Dang	Created CAD drawings of the Wind Turbine and all of the Wind Turbine components including the main shaft, half-round PVC blade 20 mm diameter, aluminium tube 60 mm length with three holes to be connected to the PVC blade, Wooden block to hold the short pipe/blades over the shaft, Bearing housing upper half, Bearing housing base through CAD on Solidworks	25/10/2024	N.D
Nhan Dang	Composed the Design Report including detailed design report, CAD drawings, and material selection, brief description of wind turbine	25/10/2024	N.D

	technology, variations, and applications		
Nhan Dang	Created the table of contents	25/10/2024	N.D
Nhan Dang	Uploaded the video to Youtube	25/10/2024	N.D
Jarrel Redoblado	Wrote Test Results	25/10/2024	J.R
Hong Nam Hoang	Wrote Modifications for improved efficiency	25/10/2024	H.H
Nathan Matthew Wijaya	Wrote Modifications for improved efficiency and DFX Princples	25/10/2024	N.W
Nhan Dang	Inserted pictures of the wind turbine and all the components	25/10/2024	N.D

Table of Contents

Design Report

Detailed Design Report

Design Report

Introduction

The Junk Rig model is a vertical-axis wind turbine designed to harness wind energy efficiently.

Bill of Materials (BOM)

1. Main Shaft (S40012): Aluminium shaft, 400 mm length, turned to 10 mm diameter for 100 mm.
2. Blades (BHM1): Half-round PVC blades, 125 mm in length, with drilled holes for mounting.
3. Blades' Connector (SPM1): Aluminium tube, 60 mm in length, with drilled holes for blade attachment.
4. Wooden Block (BM1): Mounting block for blades, drilled for shaft and blade connectors.
5. Bearing Housing (BHT & BHB): Plexi plastic blocks, upper and base halves, with holes for bearings and shaft.
6. Ball Bearing (B1020): 10 mm inner diameter, 20 mm outer diameter.
7. Bolts and Nuts (BNM4, BNM8): Various sizes for assembly.

Fabrication Processes

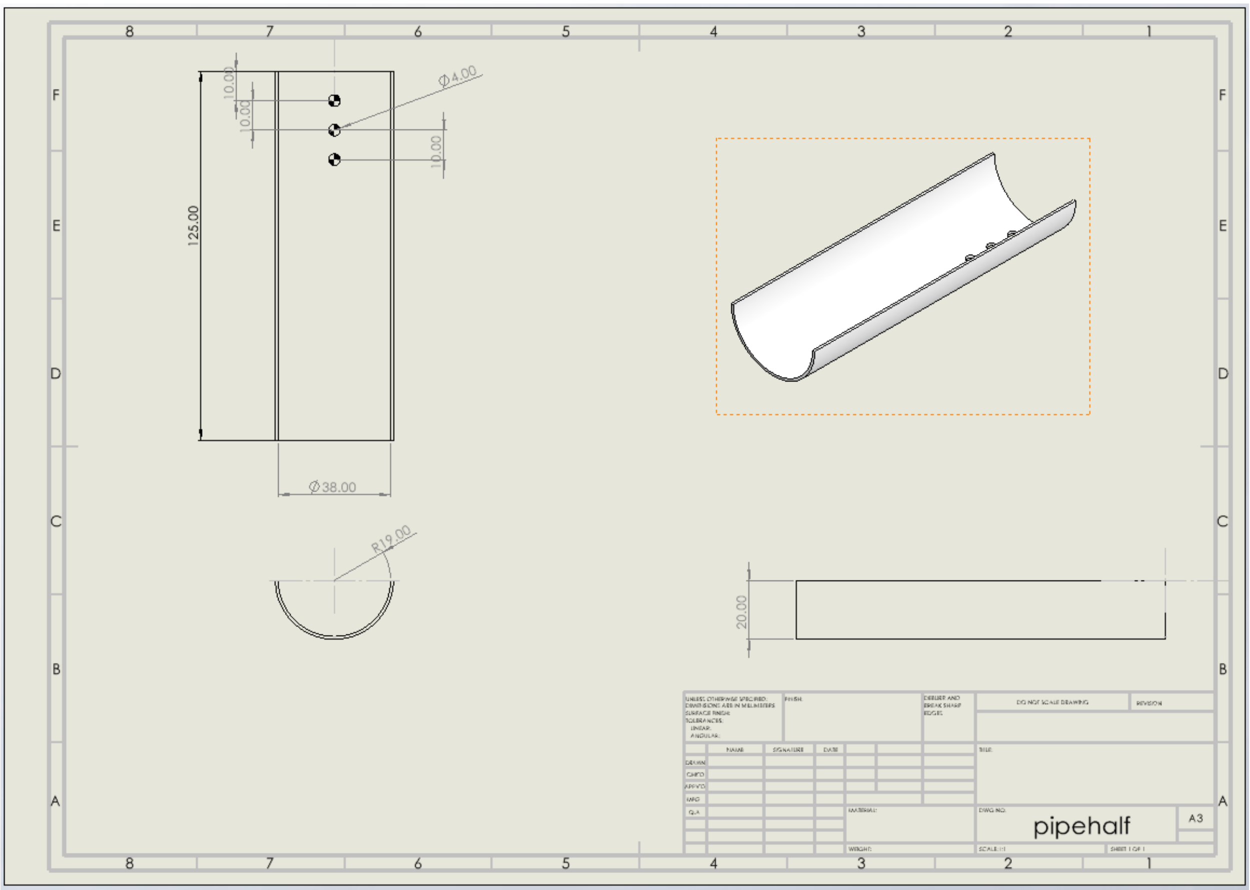
- Main Shaft: Cut and turn the aluminium bar to specified dimensions.
- Blades: Cut PVC pipe, split into half-round sections, and drill mounting holes.
- Blades' Connector: Cut aluminium tube and drill holes for blade attachment.
- Wooden Block: Cut and drill wooden block for mounting blades and shaft.
- Bearing Housing: Designed bearing housings and laser cut acrylic to make them.
- Final Assembly: Attach blades to connectors, mount on the shaft, and secure with bolts and nuts.

The Junk Rig model is a robust and efficient vertical-axis wind turbine designed for optimal performance. The detailed BOM and fabrication processes ensure precise construction and reliable operation.

CAD Drawings

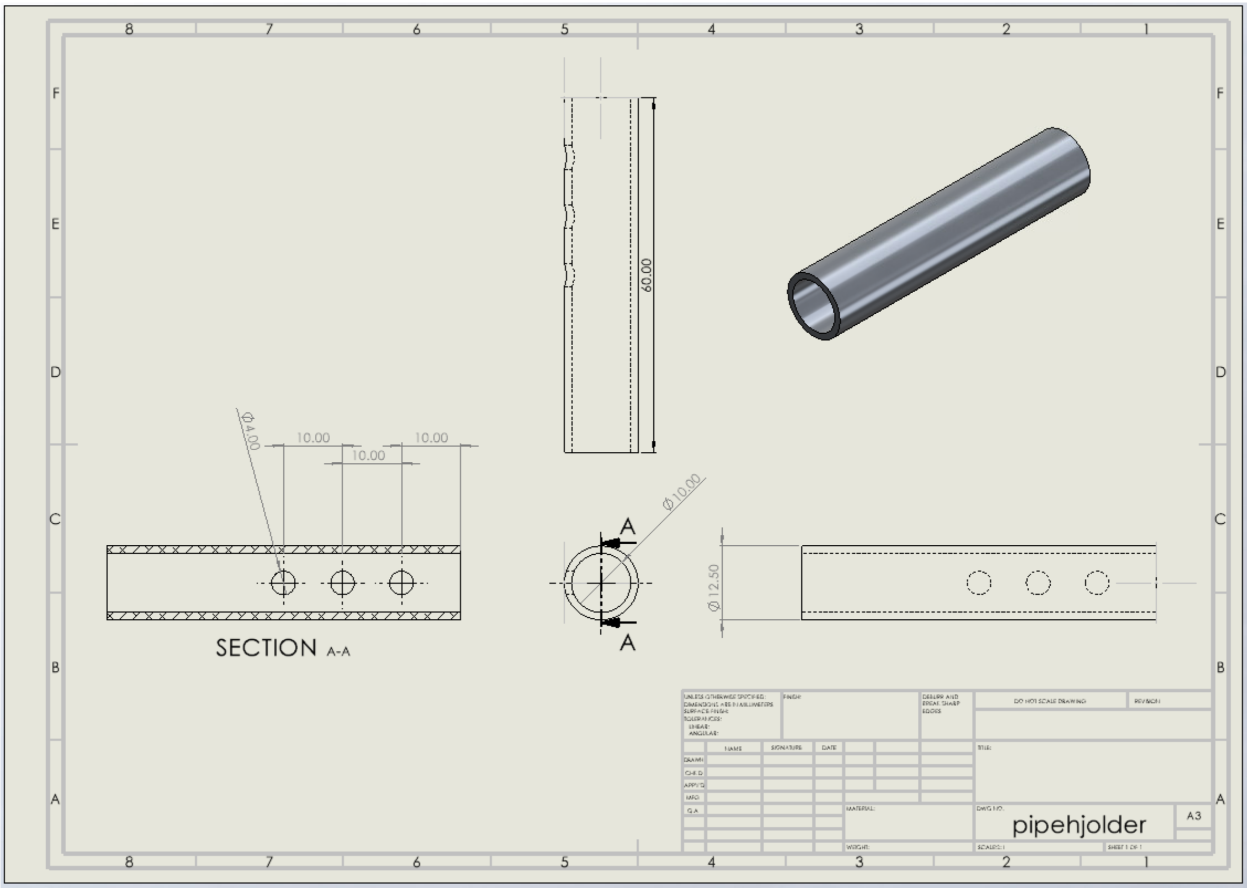
Blade

Half-Round PVC blade 20 mm diameter, the main component to harvest the mechanical power from the wind.

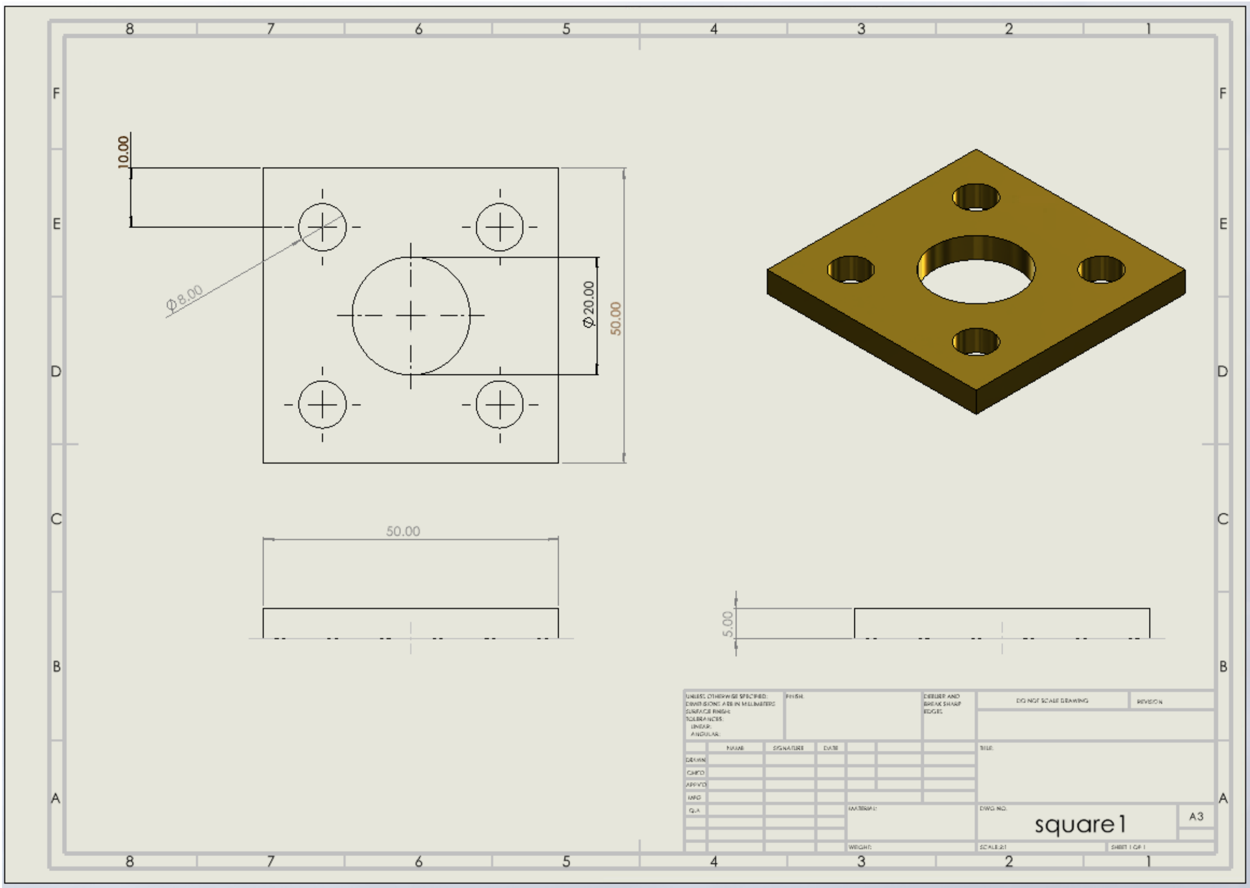


Blades’ connector

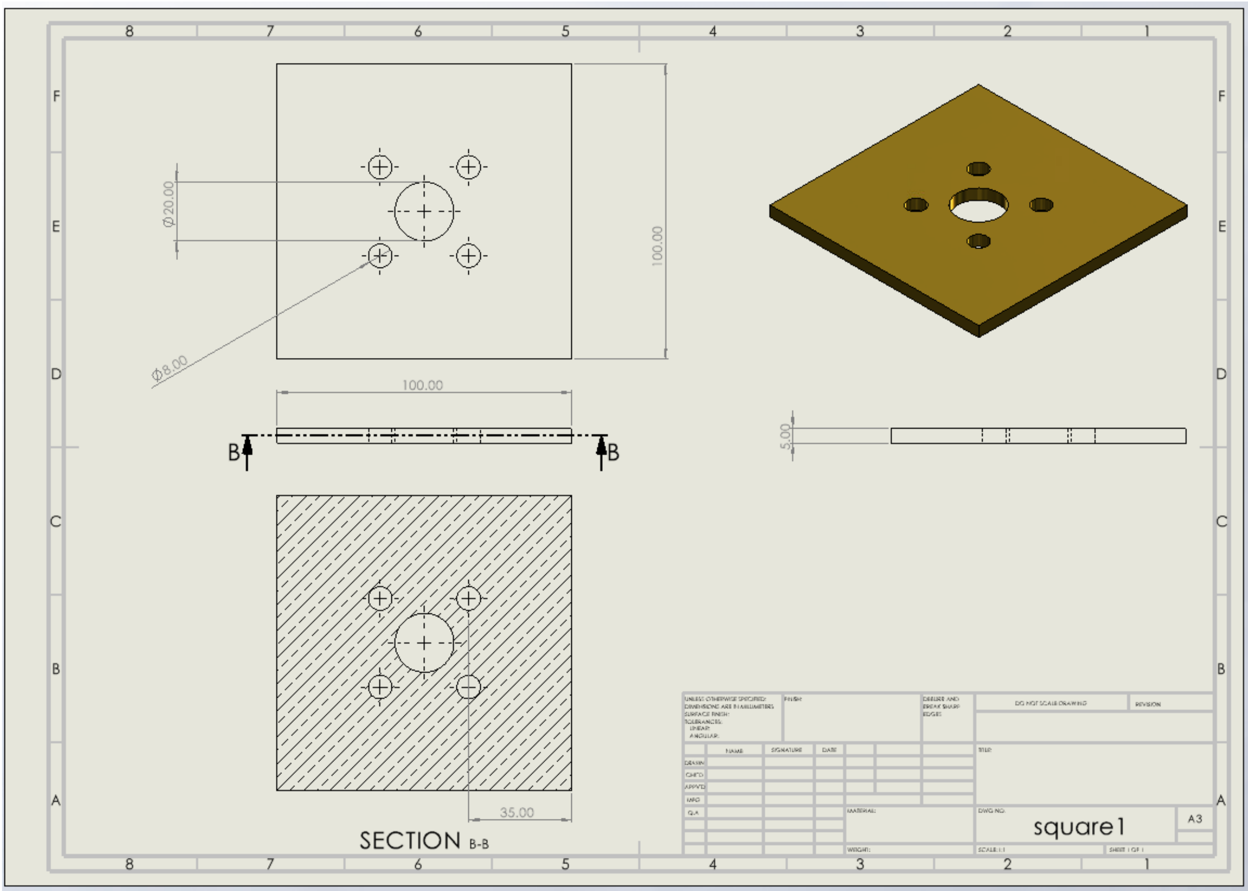
Aluminium tube 60 mm length with three holes, to be connected to the PVC blade



Bearing housing upper half

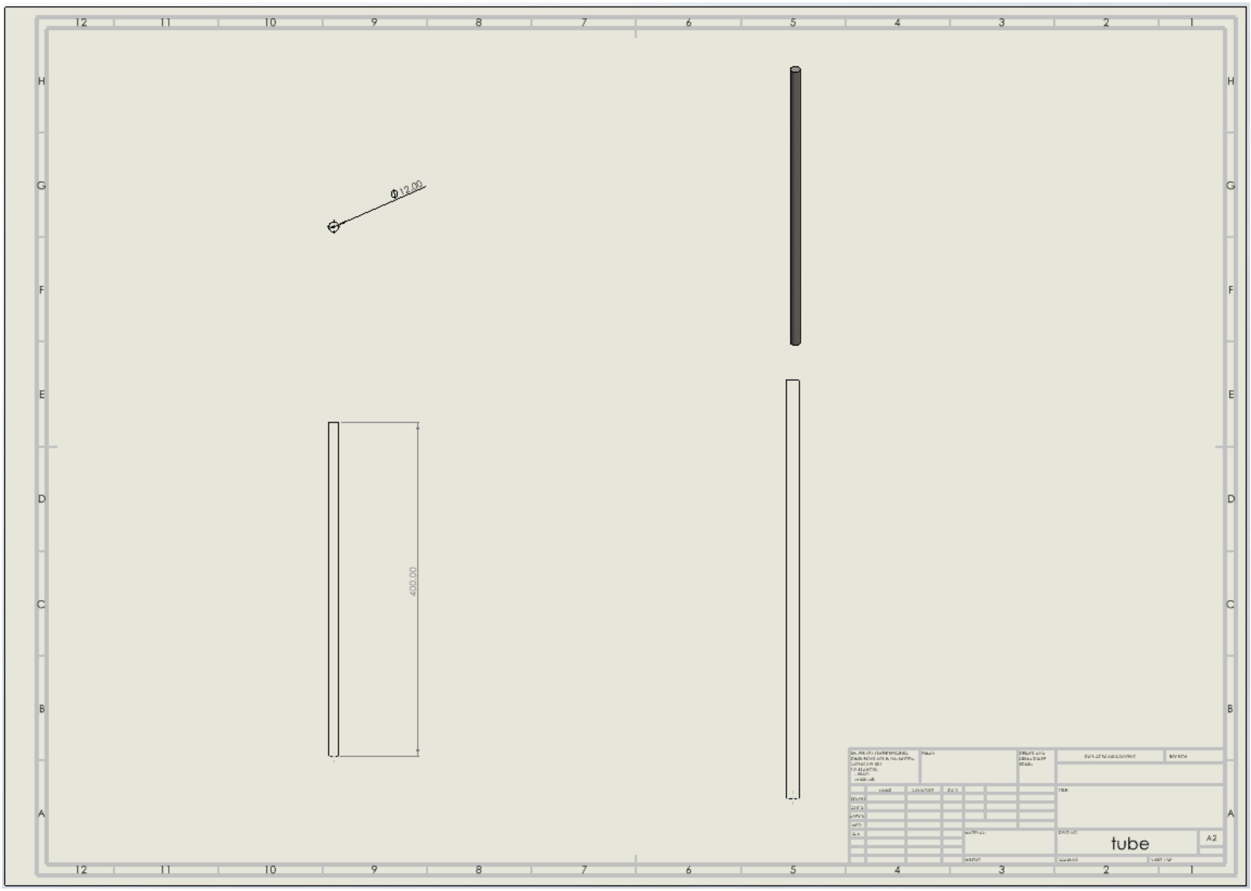


Bearing housing base



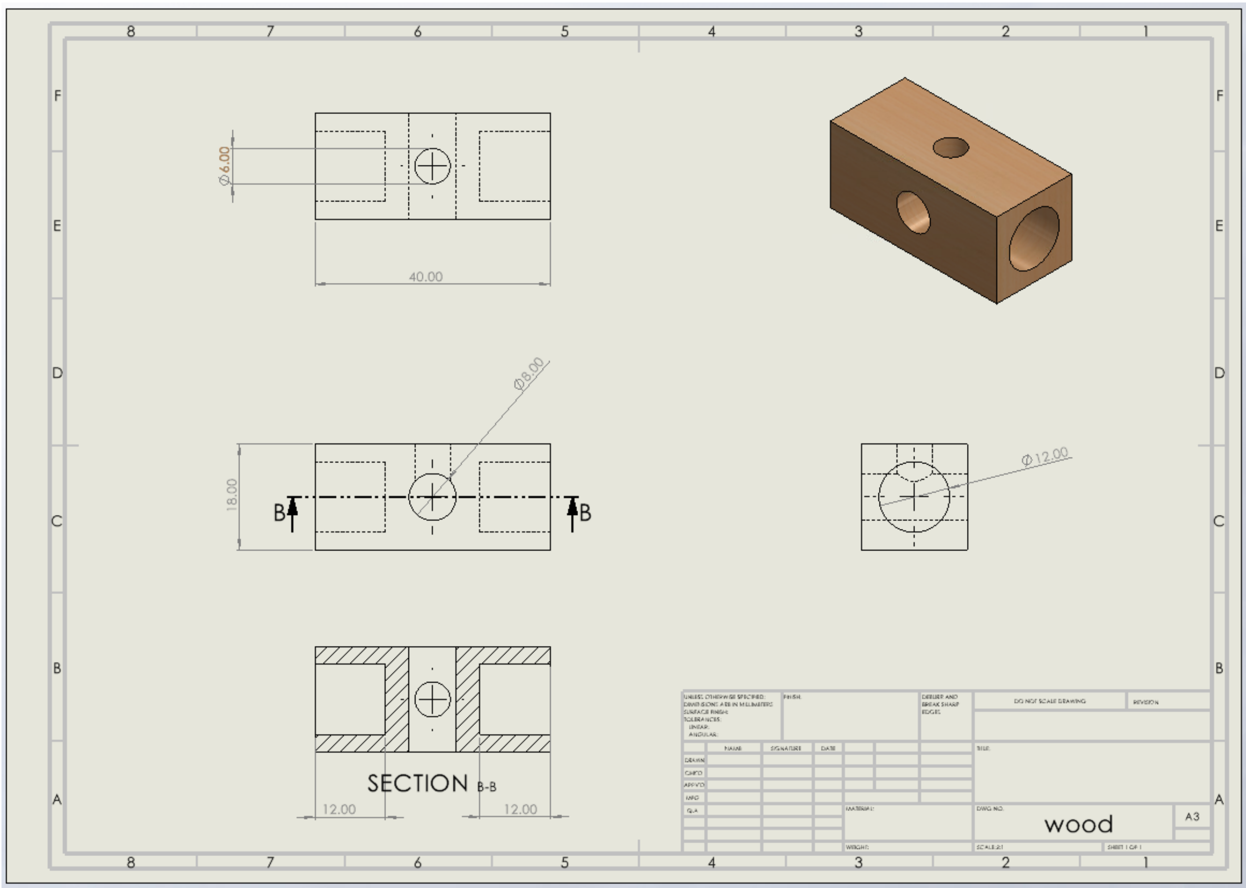
Main shaft

Aluminium shaft with two sections machined from 12 mm shaft.

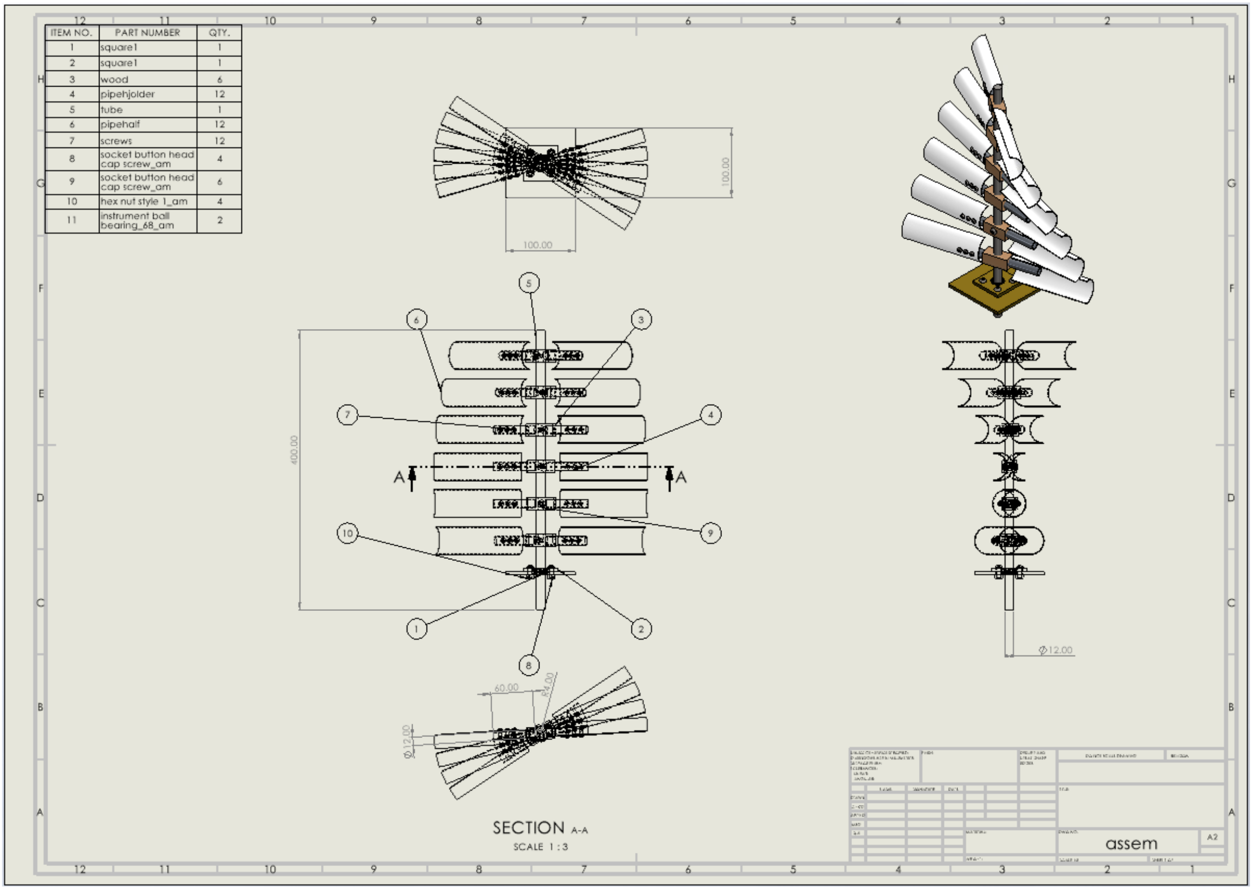


Wooden Block

Wooden block to hold the short pipe/blades over the shaft



The Junk Rig


ITEM NO.
PART NUMBER
QTY.
1 square1 1
2 square1 1
3 wood 6
4 pipehjolder 12
5 tube 1
6 pipehalf 12
7 screws 12
8 socket button head cap screw_am 4
9 socket button head cap screw_am 6
10 hex nut style 1_am 4
11 instrument ball bearing_68_am 2
100.00
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400.00
A
A
60.00
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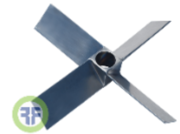
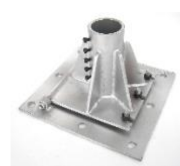
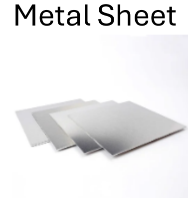
SECTION A-A
SCALE 1 : 3
assem
A2

Material Selection

Morphological Table

Parts	Function	Idea 1	Idea 2	Idea 3	Idea 4
Blade Design	Blade Shape	Half – round PVC blades 	Flat rectangular blades 	Angled Blades 	Adjustable pitch blades 
Shaft Design	Material Selection	Aluminium shaft 	Steel shaft 	Hollow aluminum shaft 	Composite shaft 
Blade Holder	Mounting method	Direct bolting to shaft	Clamping mechanism	Welded mounts	Interlocking modular design
Support Structure	Base Design	Bearing housing base 	Weighted flat base 	Anchored structure 	Floating platform
Blade Material	Material Selection	PVC 	Aluminum 	Carbon fiber 	Metal Sheet 

Justification for Material Selection

The materials chosen for the wind turbine are carefully selected to balance durability, functionality, and ease of fabrication. The main shaft is made from aluminium, a lightweight yet strong material that ensures the turbine's structural integrity while minimising weight. The blades are crafted from half-round PVC pipes, which are cost-effective and resistant to corrosion and weathering, making them ideal for outdoor use. The blade connectors, also made from aluminium, provide a robust and reliable means of attaching the blades to the shaft, ensuring stability during operation. The wooden blocks that are used to mount the blades offer a sturdy and easily machinable solution, allowing for precise alignment and secure attachment. The bearing housing components are made from Plexi plastic, chosen for its strength and ease of machining, ensuring a snug fit for the bearings and smooth rotation of the shaft. The ball bearings, with their specific inner and outer race diameters, are standard off-the-shelf items that provide low-friction rotation, essential for efficient energy conversion. Overall, the combination of these materials ensures that the vertical-axis wind turbine is both efficient and durable, capable of withstanding various environmental conditions while maintaining optimal performance.

Brief description of wind turbine technology, variations, and applications

Wind turbines are sophisticated machines designed to harness the kinetic energy of wind and convert it into electrical power. These devices typically consist of a rotor with blades, a nacelle housing the generator and gearbox, and a supporting tower. The most common configuration is the horizontal-axis wind turbine (HAWT), which dominates utility-scale applications with ratings ranging from 500 kW to 5 MW. Modern turbines employ various technologies, including fixed-speed stall-regulated systems, variable-slip designs with external rotor resistance control, and more advanced doubly-fed induction generator (DFIG) setups. The latest trend in wind turbine technology is the variable speed turbine with a full-rated power converter, offering enhanced grid integration features and independent real and reactive power control. Wind turbines can be installed both onshore and offshore, with offshore installations allowing for larger turbines and higher wind speeds. The efficiency of wind turbines is limited by Betz's law, which sets the maximum theoretical efficiency at 59.3%. Technological advancements have led to increasing turbine sizes, with some offshore models reaching up to 15 MW capacity and blade lengths exceeding 100 metres. While horizontal-axis turbines are most common, vertical-axis wind turbines (VAWTs) offer advantages in areas with highly variable wind directions. Wind turbine technology continues to evolve, with ongoing research focused on improving efficiency, reliability, and grid integration capabilities, making wind power an increasingly viable and important component of the global renewable energy mix.

Critique and Design Modifications

Modifications for improved efficiency

Current Issue: Wind turbine wasn't spinning efficiently.

Original Design Critique:

The blades were flat and did not present an optimal angle to the incoming wind. This resulted in a lack of lift generation since the angle of attack was insufficient for effective aerodynamic performance.

Modified Design:

We adjusted the angle of the blades to create an optimal angle of attack with respect to the wind. This adjustment enables the blades to interact more effectively with the wind flow, maximizing lift while minimizing drag. Specifically,

Lift vs. Drag:

- Lift Production: By angling the blades, we increased the aerodynamic lift generated by the turbine. This enhancement allows the blades to capture more wind energy, leading to greater rotational motion.
- Reduction in Drag: Non-angled blades create increased drag, leading to higher resistance against rotation. The newly positioned blades reduce drag, allowing for smoother and more efficient spinning of the rotor.

Impact on Starting Torque:

- Torque Generation: Non-angled blades tend to produce poor starting torque, making it difficult for the turbine to begin rotating, particularly in low wind conditions. The new angled blades improve the torque generated, even at low wind speeds, by presenting a larger surface area at an optimal angle.
- Cut-In Wind Speed: This modification effectively lowers the cut-in wind speed—the minimum wind speed at which the turbine starts to generate power. The ability to initiate rotation at lower wind speeds enhances overall efficiency and energy capture, as the turbine can begin producing power sooner in varied wind conditions.

The lift force can be calculated using the lift equation:

$$F_L = \frac{1}{2} \cdot \rho \cdot V^2 \cdot A \cdot C_L$$

- F_L = Lift force (N)
- ρ = Air density (kg/m^3)
- V = Wind speed (m/s)
- A = Cross-sectional area of the blade (m^2)
- C_L = Lift coefficient (dimensionless), which depends on the angle of attack.

The formula above is provident that how the angle of the blades and the area affects the lift force.

Besides, we realized that the engine oil is also helpful to reduce the friction between the shaft (aluminum) and the base. Specifically,

- **Friction Issues**: The initial design faced challenges related to friction between the aluminum shaft and its base. Excessive friction can hinder the rotational speed and efficiency of the turbine, leading to energy losses.
- **Lubrication Modification**: To mitigate this issue, we introduced engine oil as a lubricant between the shaft and the base. The oil serves to reduce friction effectively, ensuring smoother operation of the turbine.

Finally, to address stability concerns, we observed that the blades were prone to movement due to structural issues within the turbine's assembly. Without secure fastening, the blades are at risk of detachment or misalignment, especially under higher wind speeds, which compromises the efficiency and durability of the turbine. To enhance structural stability, we decided to apply silicone adhesive to bond the blades firmly to the wooden cube and the central shaft. Silicone was chosen for its strong adhesion properties, flexibility, and weather-resistant qualities, ensuring that the blades remain securely in place despite the constant wind force.

Critiques and Proposed Modifications

The recent designs have some issues regarding the blades not spinning properly. The design maybe simple and easy to be manufactured, but it doesn't ignore that there are some issues regarding the blades. The blades angle were not optimal and it causes the problem. We adjusted the angle of the blades to create an optimal angle of attack with respect to the wind. With this adjustment, it enables the blades to interact more efficiently and work more optimally, maximizing lift while minimizing drag and this solves the issues and makes the performance of the turbine overall better. Another modification is the number of blades were increased, from 8 to 10. It will improve the performance overall, because with more surface area this can extract more wind energy and generates more torque. Also with more blades the turbine has a lower cut-in speed, which means it can start to generate power at lower wind speeds.

Integration of DFX principles in redesign

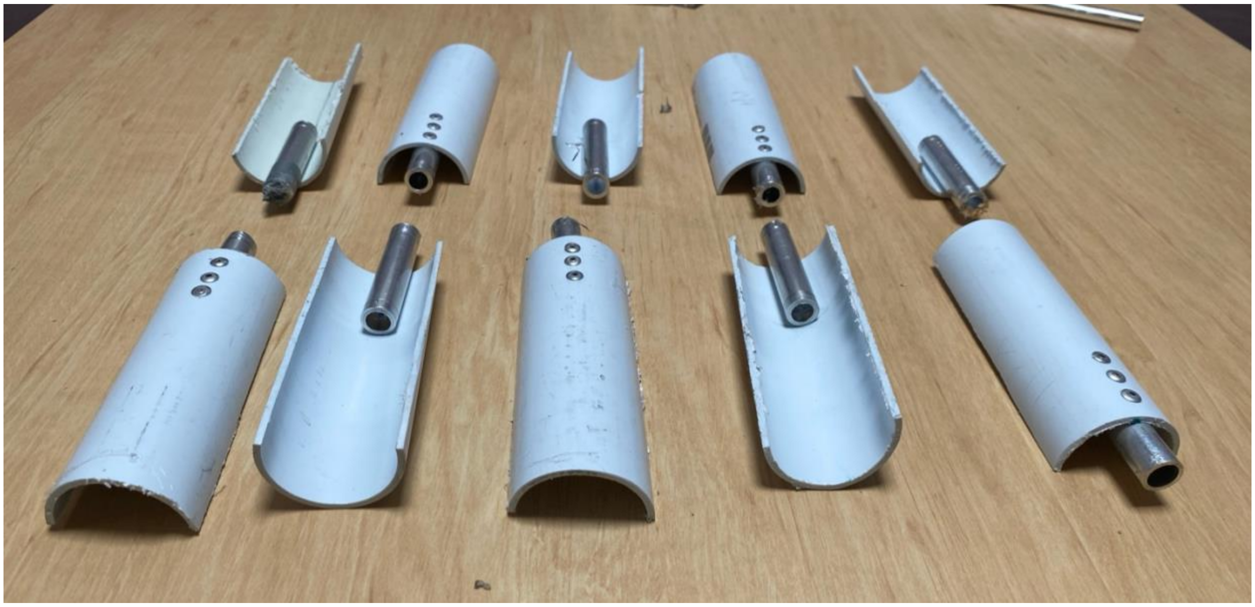
The integration of DFX principles that are used in the redesign are, design for performance and design for manufacturability. By changing the angle of the blades and increasing the number of blades, it improves the overall ability of the turbine. By maximizing the lift and drag, the modification ensures that the blades work more efficiently resulting in incresed performance. This aligns with design for performance principle. The second principle that is integrated to the redesign is design for manufacturability. By changing the angle and adding the number of blades to improve the performance, with this modification changed without significantly complicating the design or the manufacturing process. The changes that were applied to the turbine enhances the performance while still maintaining a design that is not significantly different from the first design.

Fabricated Turbine

Wind Turbine

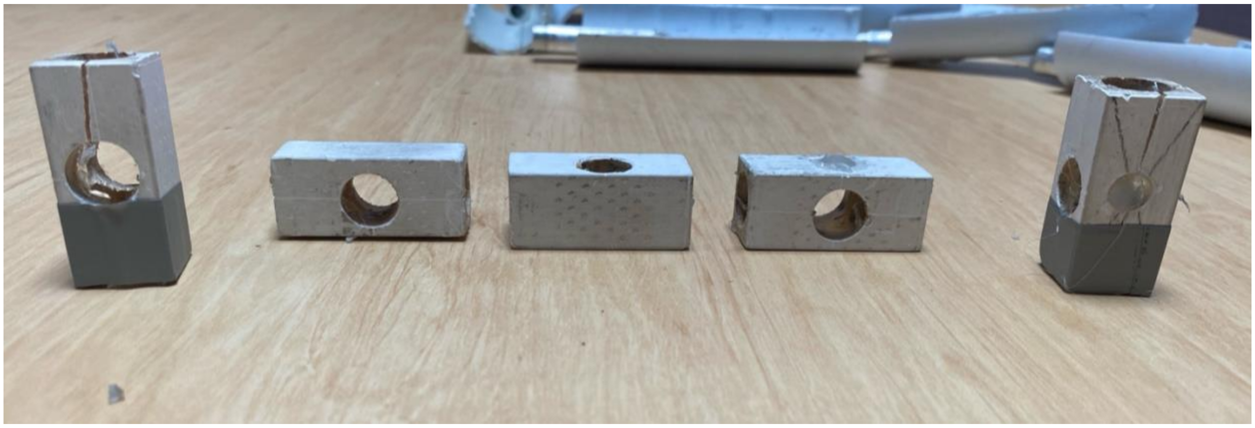
Main shaft

Blades



Blades’ connectors

Wooden Blocks



Bearing housing upper half and bearing housing base

The bearing housing upper half and the bearing house base were laser cut. However, another housing base was provided in the wind tunnel.

Test Results

Performance testing and reporting

Objective:

The objective of the testing was to evaluate the performance of the wind turbine in terms of efficiency, structural integrity, and stability under various controlled wind speeds.

Testing Setup:

Wind Speed Range: 5 m/s

Environment: Indoor testing facility (wind tunnel)

Test Procedures:

1. Place wind turbine on housing bearing base
2. Turn on wind tunnel
3. Write down any observations

Structural Observations:

PVC blades performed adequately but exhibited flexibility that contributed to minor stability issues at higher speeds. For improved high-speed performance, a stiffer blade material (e.g., carbon fiber) may reduce blade flex and increase power output.

Blade motion

Efficiency peaked at wind speeds of 6-7 m/s, with a noticeable decline as speeds reached 8-10 m/s. High-speed operation seemed to create a “stall” effect on the PVC blades, causing a drop in rotational efficiency. Efficiency at lower wind speeds was marginal due to slower blade rotation, indicating that the turbine requires a minimum threshold wind speed (~5 m/s) for effective performance.

Structural Stability

Structural vibrations were minimal. The turbine operated smoothly without any significant wobbling or lateral movement.

Video Link of Lab Test

https://youtu.be/HkvrupotTRQ